

POST-SILICON TESTING

Considering that every chip in the world and every transistor on the chip are tested, just imagine the amount of time and cost involved in testing millions of chips with billions of gates. Semiconductor manufacturers are always looking for ways to reduce the test time and cost

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Even if you use all the latest tools for your pre-silicon verification and develop the most accurate model near to the actual system, the biggest challenge is to get the first working silicon," claims Rajamohan Varambally, director-technology R&D, STMicroelectronics, India.

The primary goal of post-silicon testing or validation is to ensure that the intent of your design is met. Varambally adds, "Making a chip for the first time will cost several million dollars. This is why whatever can be done with software is done beforehand so that the chances of coming up with correct silicon first time itself are high. However, software cannot simulate the entire working. Final testing has to be done on silicon."

Silicon design and manufacturing processes have evolved over the years. Neeraj Rai, verification signoff manager, ASIC division, elfnchip, says, "The industry is dynamic and the methodology and languages have changed. The level of abstraction is a lot higher today. To support this, tool vendors and test equipment providers have developed more efficient and automated test systems."

Post-silicon validation or testing has two aspects. Naresh Kumar, general manager, applications (EMG), Agilent Technologies, explains, "Once the first silicon is made, it is tested for functionality using discrete equipment

or configured automated test equipment (ATE) before sending for mass production. Every chip produced is then tested using large testers." In functional test, fault-finding or debugging is critical, while in production time-to-test is critical.

SoC: Need for embedded multi-functional tester

As every chip is designed for different purpose, the test parameters and process for functional test vary from chip to chip. Testing a chip for its functionality is basically providing a stimulus and measuring the response. There are advanced test systems like the ATEs which are configured to perform test

on a particular chip.

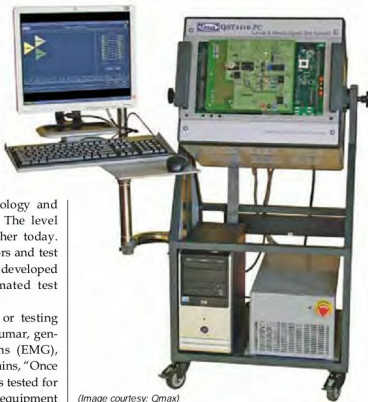
"Once the designer provides specifications of the integrated circuit (IC), test programs are written to test its functionality. The program will feed in the inputs and measure the corresponding output to validate the functionality as per the specifications," explains Veerappan V., co-founder and president-operations, Tessolve.

Rai adds "A test program written to test an analogue-to-digital converter (ADC) will be much different from the test program written for testing MP3 decoder chip." It is very important to correctly program the ATE to do the functional testing.

There are analogue, digital and mixed test systems to test various parameters of the chips. The development of system-on-a-chip (SoC) demands testing of multiple functions simultaneously. The biggest challenge here is the development of multifunctional test programs. Embedded multifunctional test is of increasing importance in enabling SoC technology. There are software tools that help in test program conversion and enable smooth multifunctional tests.

Characterisation and reliability tests

Apart from testing for the actual functionality,



(Image courtesy: Qmax)